

Sarah Kane
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EDUCATION

University of Pennsylvania, Philadelphia, PA
Bachelor of Arts (*in progress*), Physics (Astrophysics), Expected Graduation: May 2023
GPA: 4.0

West Morris Mendham High School, Mendham, NJ
(September 2015 - June 2019), Final Weighted GPA: 5.814

Relevant Coursework: Computational Physics, Introduction to Astrophysics I & II, Data Analysis for the Natural Sciences, Machine Learning, Topics in Cosmology, Galaxy Structure & Formation, Analytical Mechanics (*in progress*), Electromagnetism I & II (*in progress*)

HONORS AND AWARDS

Thomas H. Wood Prize, University of Pennsylvania Department of Physics and Astronomy (2020)

“Awarded annually to the undergraduate student in introductory physics who has demonstrated, during the course of their undergrad coursework, the greatest proficiency in assimilating the concepts of physics”

Penn Undergraduate Research Mentorship Program (PURM), University of Pennsylvania Center for Undergraduate Research and Fellowships (2020)

“PURM provides students completing their first or second undergraduate year the opportunity to spend a summer conducting cutting-edge research with a standing Penn faculty member.”

Valedictorian, West Morris Mendham High School (2019)

National Merit Scholarship Finalist (2019)

National AP Scholar (2019)

Mayo Performing Arts Center Scholarship (2019)

Mendham Township Recreation: Patriots Community Service Award (2019)

Mendham High School Drama Award for Excellence in Theater (2019)

McGowan Leadership Scholarship, Honorable Mention (2019)

RESEARCH EXPERIENCE

Research Assistant to Professor Bhuvnesh Jain

University of Pennsylvania, Department of Physics and Astronomy, May 2020 - Present
Developed a machine learning algorithm to search for exoplanet transits in data from NASA's Transiting Exoplanet Survey Satellite (TESS)
Studied star formation rate quenching of galaxies infalling into clusters

Usability Tester for Astronify

National Aeronautics and Space Administration (NASA), 2020

Participated in usability tests for Astronify, a NASA project designed to convert time series astronomical data into sound, particularly to improve the accessibility of this data for the blind and visually impaired

<https://astronify.readthedocs.io/en/latest/>

PRESENTATIONS

“Planetary Systems Around Milky Way Stars,” Poster Presentation at Penn Undergraduate Research Fair, 2020

“Searching for Exoplanets with Machine Learning,” Poster Presentation at Penn Undergraduate Research Fair, 2021

Featured in “Disability Advocacy and the Sciences” in *OMNIA*, the publication for the University of Pennsylvania’s Arts and Sciences, 2022

OUTREACH

Moelis Access Science Fellow

The Netter Center for Community Partnerships, University of Pennsylvania

October 2019 - Present

Provide students in Philadelphia after-school science enrichment programs to improve the quality of the science curriculum in city schools

Volunteer to help run the Science Olympiad Invitational competition for high school students at the University of Pennsylvania

Research Peer Advisor

Center for Undergraduate Research and Fellowships, University of Pennsylvania

September 2020 - Present

Advise Penn undergraduate students on how to prepare for and find research positions, particularly in physics and astronomy

Hosted a presentation on finding research positions for Penn students with disabilities

National Organization for Albinism and Hypopigmentation (NOAH)

Hosted “NOAH Connections” Zoom calls for people with albinism across the country who have or want to pursue careers in STEM, April 2021 - July 2021 - see albinism.org for details

Field Trip from the Overbrook School for the Blind

December 2021

Organized a field trip for students from the Overbrook School for the Blind to the University of Pennsylvania’s Department of Physics and Astronomy. The students performed a lab activity on sound waves and heard talks from professors on subjects ranging from black holes and gravity to how telescopes work.

Tutoring

February 2018 - Present

Work as a private tutor for high school & college students in physics, chemistry, biology, and math

MEMBERSHIP

Sonification World Chat

A monthly call for scientists, data experts, and musicians around the world who have an interest in turning scientific data into sound, especially for accessibility purposes for the blind and visually impaired

Women in Physics at the University of Pennsylvania

Diversity and Inclusion in Physics at the University of Pennsylvania

Society of Physics Students at the University of Pennsylvania

Disability Advocacy @ Penn

SKILLS

Computational: Python (including Numpy, Scipy, Pandas, Matplotlib), Machine Learning using Tensorflow and Scikit-Learn, experience working with a variety of astronomical data programmatically (including data from the Transiting Exoplanet Survey Satellite, Illustris-TNG, CosmoSim, and the Sloan Digital Sky Survey)

Mathematics and Statistics: Calculus (including multivariable), Linear Algebra, Differential Equations, Regression Models, Principal Component Analysis, Decision Trees, Random Forest, Linear and Quadratic Discriminant Analysis, Resampling Methods